

Minh Canh Vu, PhD

Polymer/Electrochemical/Material Scientist

Professional Appointments

Assistant Professor VinUniversity	Hanoi, Vietnam 06/2026 – now
Research Associate The University of Newcastle	Newcastle, Australia 06/2024 – 05/2026
Postdoctoral Fellow with Prof. Chibueze V. Amanchukwu The University of Chicago, Pritzker School of Molecular Engineering	Chicago, USA 12/2021 – 05/2024
Postdoctoral Researcher with Prof. Sung-Ryong Kim Korea National University, Polymer Science and Engineering	Chungju, South of Korea 03/2021 – 12/2021

Education

Korea National University Ph.D. in Polymer Science and Engineering <i>Thesis:</i> Graphene fluoride for thermoconductive nanocomposites	Chungju, South of Korea 2015 – 2020
Industrial University of Ho Chi Minh City B.S. in Chemical Engineering <i>Thesis:</i> Green synthesis of lubricant oil from oleic acid	Ho Chinh Minh City, Vietnam 2010 – 2014

Granted Fundings

1. Australia-USA Battery Exchange Program, **Chief-Investigator (CI)**, 2025. Grant: \$30,000.
2. University of Newcastle, PILOT Scheme Program, **CI**. Grant: \$15,000.
3. Development of thermally conductive electromagnetic interference shielding materials based on 2D materials for 5G electronics, National Foundation for Science and Technology Development, **CI**. Grant: ~\$94,000.
4. Development of stretchable thermal interfacial materials based on graphene fluoride materials, Vietnamese Ministry of Education and Training, **co-CI**. Grant: ~\$38,000.
5. Enabling energy-dense grid scale batteries with earth abundant materials, Department of Energy, **Participant Investigator (PI)**, 2022-2025. Grant: ~\$500,000.
6. Development of thermally conductive and yet electrically insulating pressure sensitive adhesives using 3D nanostructures, Korean Ministry of Education, **PI**, 2017-2022. Grant: ~\$387,000.
7. Development of thermally conductive film and bonding materials for display devices with in-plane thermal conductivity of 300 W/mK, Korean Ministry of Trade, Industry & Energy, **PI**. Grant: ~\$332,000.
8. Development of solvent-free adhesives with a peel strength of 3500 g/inch and property change with a timeless than 10% for electronic applications and processing technology using thereof, Korean Ministry of Trade, Industry & Energy, **PI**, 2020-2022. Grant: ~\$210,000.
9. Development of stretchable thermally conductive pressure-sensitive adhesives for flexible display, Korean Ministry of SMEs and Startups, **PI**, 2018-2019. Grant: ~\$110,000.
10. Development of carrier-less flexible adhesive films for curved display applications with superhydrophobic contact angle larger than 100-degree, Korean Ministry of SMEs and Startups, **PI**, 2018-2019. Grant: ~\$110,000.
11. Development of transparent medical patch using nanostructures, Korean Ministry of Education, **PI**, 2016-2019. Grant: ~\$165,000.

Publications

[#]Equal contribution *Corresponding author



Patents

1. S.R. Kim, **M.C. Vu**, Y.M. Kang, Surface modified carbon flakes and the carbon brush motor using thereof, granted *Korean Patent 10-2018-0028868*, 2020.
2. S.R. Kim, **M.C. Vu**, Manufacturing method of SiC sheet and thermally conductive composites using thereof, granted *Korea Patent 10-2018-0077307*, 2020.
3. S.R. Kim, **M.C. Vu**, Multi-functional bioinspired paper for thermal management and electromagnetic interference shielding, method of fabrication for the same, granted *Korean Patent 10-2020-0124849*, 2020.
4. S.R. Kim, **M.C. Vu**, Hybrid heat-radiating film and its manufacturing method, granted *Korean Patent 10-2020-0100099*, 2020.
5. S.R. Kim, **M.C. Vu**, Development of paper with aramid nanofiber/graphene fluoride/aramid nanofiber structure

- with excellent mass production and high-temperature capacitive energy storage capability, granted Korea patent, 2021.
6. S.R. Kim, **M.C. Vu**, High heat-dissipating electromagnetic wave shielding film manufactured by cross-linking graphene nanoplate and aramid nanofiber and its manufacturing method, granted Korean patent, 2021.
 7. S.R. Kim, **M.C. Vu**, Hybrid thermal management film and method of fabrication for the same, granted US20220041912A1, 2022.
 8. S.R. Kim, **M.C. Vu**, Energy storage film and method of manufacturing same, granted US20230127153A1, 2023.
 9. C.V. Amanchukwu, **M.C. Vu**, H. Srivinasan, A transition-metal-and-solvent-free dual-graphite battery for long duration storage enabled by a fluorosulfonyl imide-based molten salt electrolyte, filed US 25-T-043.

Corresponding Author Publications

1. TH Nguyen, CT Nguyen, HA Le Pham, VC Nguyen, **M.C. Vu***, Flexible Thermal Conductive Electromagnetic Interference Shielding of Liquid Metal-Based Silicon Polymer Composites: Strain Level Effect, *J Appl Polym Sci*, (2025) e70175. (**Q2, H-index 200, IF 2.8**)
2. V.C. Nguyen, T.L. Chau, C.T. Nguyen, H.A.L. Pham, S. Ghotekar, **M.C. Vu***, Flexible Thermal Conductive Electromagnetic Interference Shielding of Liquid Metal-Based Silicon Polymer Composites: Strain Level Effect, *J Appl Polym Sci*, **42** (2025) e57243. (**Q2, H-index 200, IF 2.8**)
3. D.K. Nguyen, T.N. Pham, A.L.H. Pham, V.C. Nguyen, M.S. Tran, V.Q. Bui, **M.C. Vu**, Multilayered silver nanowires and graphene fluoride-based aramid nanofibers for excellent thermoconductive electromagnetic interference shielding materials with low-reflection, *Colloids Surf A Physicochem Eng Asp* **688** (2024) 133553. (**Q1, H-index 205, IF 5.4, citation: 8**)
4. T.C. Nguyen, A.L.H. Pham, D.K. Nguyen, M.T.K. Lien, V.C. Nguyen, **M.C. Vu**, Ultrathin Aramid Nanofiber Composites with Alternating Multilayered Structure of Silver Nanowires and Boron Arsenide: Toward Superior Electrically Insulating Thermoconductive Electromagnetic Interference Shielding Materials, *ACS Appl Electron Mater* **6** (2024) 3704–3716. (**Q1, H-index 64, IF 4.7, citation: 9**)
5. V. C. Nguyen*, V. N. Pham, C. T. Nguyen, H.A.L. Pham, D.K. Nguyen, M.T.K. Lien, **M.C. Vu**, Layer-by-layer assembly of boron arsenide and copper nanoflake-based aramid nanofibers for thermoconductive electromagnetic interference shielding materials with superior mechanical flexibility and flame retardancy, *Journal of Industrial and Engineering Chemistry* **138** (2024) 492-501. (**Q1, H-index 153, IF 5.9, citation: 4**)
6. D.K. Nguyen, T.T.H. Tran, T.K.L. Mai, M.S. Tran, S. Ghotekar, A.L.H. Pham, V.C. Nguyen*, **M.C. Vu**, Thermoconductive Graphene Fluoride Cross-Linked Aramid Nanofiber Composite Films with Enhanced Mechanical Flexibility and Flammable Retardancy for Thermal Management in Wearable Electronics, *ACS Appl Nano Mater* **7** (2024) 2724–2734. (**Q1, H-index 98, IF 5.5, citation: 4**)
7. D.K. Nguyen, H.N. Truong, A.L.H. Pham, M.S. Tran, M.T. Nguyen Dinh, V.Q. Bui, M.T.K. Lien, V.C. Nguyen*, **M.C. Vu**, Ultratough and self-healable electromagnetic interference shielding materials with sandwiched silver nanowires in polyurethane composite films, *Polym Compos* **45** (2024) 14943-14952. (**Q1, IF 4.7, citation: 5**)
8. T.T.V. Tran, D.V.N. Vo, S.T. Nguyen, **C.M. Vu**, Silver nanowires decorated recycled cigarette filters-based epoxy composites with high through-plane thermal conductivity and efficient electromagnetic interference shielding, *Compos Part A Appl Sci Manuf* **149** (2021) 106485. (**Q1, H-index 227, Ranking 13/130 in Ceramics and Composites, IF 8.9, 35 citations,**)
9. T.T.V. Tran, V.A. Le, C.H. Nguyen, D.V.N. Vo, T.M. Nguyen, C. Samart, S. Kongparakul, S. Ghotekar, **C.M. Vu**, Thermoconductive Ultralong Copper Nanowire-Based Aramid Nanofiber Nanopapers for Electromagnetic Interference Shielding, *ACS Appl Nano Mater* **5** (2022) 12991–13001. (**Q1, H-index 98, IF 5.5, 22 citations**)
10. T.T.V. Tran, D.V.N. Vo, S.T. Nguyen, S.D.N. Luu, M. Mofijur, **C.M. Vu**, In-situ sintered silver decorated 3D structure of cellulose scaffold for highly thermoconductive electromagnetic interference shielding epoxy nanocomposites, *J Appl Polym Sci* **138** (2021) 51193. (**Q2, H-index 200, IF 2.8, 35 citations**)

First Author Publications

1. D. Mani, **M.C. Vu[‡]**, S. Anand, J.B. Kim, T.H. Jeong, I.H. Kim, B.K. Seo, M.A. Islam, S.R. Kim, Elongated liquid metal based self-healing polyurethane composites for tunable thermal conductivity and electromagnetic interference shielding, *Composites Communications* **44** (2023) 101735. (**Q1, H-index 73, Ranking Top 10% in Polymers and Plastics, IF 7.7, 52 citations**)
2. **M.C. Vu**, P. Mirmira, R.J. Gomes, P. Ma, E.S. Doyle, H.S. Srinivasan, C. V. Amanchukwu, Low melting alkali-based molten salt electrolytes for solvent-free lithium-metal batteries, *Matter* **6** (2023) 4357–4375. (**Q1, H-index 111, Ranking 13/360 in Materials Science, IF 17.5, 23 citations**)
3. **M.C. Vu**, D. Mani, T.H. Jeong, J.B. Kim, C.S. Lim, H. Kang, M.A. Islam, O.C. Lee, P.J. Park, S.R. Kim, Nacre-

- inspired nanocomposite papers of graphene fluoride integrated 3D aramid nanofibers towards heat-dissipating applications, *Chemical Engineering Journal* **429** (2022) 132182. (Q1, H-index 337, Ranking 7/274 in Chemical Engineering, IF 13.2, 57 citations)
4. **M.C. Vu**, H. Kang, P.J. Park, B.G. Choi, J.W. Paik, W.K. Choi, M.A. Islam, Q. Wang, S.R. Kim, Scalable graphene fluoride sandwiched aramid nanofiber paper with superior high-temperature capacitive energy storage, *Chemical Engineering Journal* **444** (2022) 136504. (Q1, H-index 337, Ranking 7/274 in Chemical Engineering, IF 13.2, 27 citations)
 5. **M.C. Vu**, P.J. Park, S.R. Bae, S.Y. Kim, Y.M. Kang, W.K. Choi, M.A. Islam, J.C. Won, M. Park, S.R. Kim, Scalable ultrarobust thermoconductive nonflammable bioinspired papers of graphene nanoplatelet crosslinked aramid nanofibers for thermal management and electromagnetic shielding, *J Mater Chem A Mater* **9** (2021) 8527–8540. (Q1, H-index 318, Ranking 32/460 in Materials Science, IF 9.5, 88 citations)
 6. **M.C. Vu**, D. Mani, J.B. Kim, T.H. Jeong, S. Park, G. Murali, I. In, J.C. Won, D. Losic, C.S. Lim, S.R. Kim, Hybrid shell of MXene and reduced graphene oxide assembled on PMMA bead core towards tunable thermoconductive and EMI shielding nanocomposites, *Compos Part A Appl Sci Manuf* **149** (2021) 106574. (Q1, H-index 227, Ranking 13/130 in Ceramics and Composites, IF 8.9, 105 citations)
 7. **M.C. Vu**, T.H. Jeong, J.B. Kim, W.K. Choi, D.H. Kim, S.R. Kim, 3D printing of copper particles and poly(methyl methacrylate) beads containing poly(lactic acid) composites for enhancing thermomechanical properties, *J Appl Polym Sci* **138** (2021) 49776. (Q2, H-index 200, IF 2.8, 36 citations)
 8. **M.C. Vu**, W.K. Choi, S.G. Lee, P.J. Park, D.H. Kim, M.A. Islam, S.R. Kim, High Thermal Conductivity Enhancement of Polymer Composites with Vertically Aligned Silicon Carbide Sheet Scaffolds, *ACS Appl Mater Interfaces* **12** (2020) 23388–23398. (92 citations, Q1, H-index 334, Ranking Top 10% (14/460) in Materials Science, IF 8.5, 95 citations)
 9. **M.C. Vu**, N.A. Thi Thieu, J.H. Lim, W.K. Choi, J. Chan Won, M.A. Islam, S.R. Kim, Ultrathin thermally conductive yet electrically insulating exfoliated graphene fluoride film for high performance heat dissipation, *Carbon* **157** (2020) 741–749. (106 citations, Q1, H-index 346, Ranking 19/460 in Chemistry, IF 11.6, 112 citations)
 10. **M.C. Vu**, N.A.T. Thieu, W.K. Choi, M.A. Islam, S.R. Kim, Ultralight covalently interconnected silicon carbide aerofoam for high performance thermally conductive epoxy composites, *Compos Part A Appl Sci Manuf* **138** (2020) 106028. (Q1, H-index 227, Ranking 13/130 in Ceramics and Composites, IF 8.9, 36 citations)
 11. **M.C. Vu**, I.H. Kim, W.K. Choi, C.S. Lim, M.A. Islam, S.R. Kim, Highly Flexible Graphene Derivative Hybrid Film: An Outstanding Nonflammable Thermally Conductive yet Electrically Insulating Material for Efficient Thermal Management, *ACS Appl Mater Interfaces* **12** (2020) 26413–26423. (Q1, H-index 334, Ranking Top 10% (14/460) in Materials Science IF 8.5, 57 citations)
 12. **M.C. Vu**, Q.V. Bach, D.D. Nguyen, T.S. Tran, M. Goodarzi, 3D interconnected structure of poly(methyl methacrylate) microbeads coated with copper nanoparticles for highly thermal conductive epoxy composites, *Compos B Eng* **175** (2019) 107105–107113. (Q1, H-index 227, Ranking 11/720 in Mechanical Engineering, IF 14.2, 46 citations)
 13. **M.C. Vu**, Y.H. Bae, M.J. Yu, M.A. Islam, S.R. Kim, Core-shell structured carbon nanotube-poly(methylmethacrylate) beads as thermo-conductive filler in epoxy composites, *Compos Part A Appl Sci Manuf* **109** (2018) 55–62. (Q1, H-index 227, Ranking 13/130 in Ceramics and Composites, IF 8.9, 26 citations)
 14. **M.C. Vu**, T.S. Tran, Y.H. Bae, M.J. Yu, V.C. Doan, J.H. Lee, T.K. An, S.R. Kim, Self-Assembly of Carbon Nanotubes and Boron Nitride via Electrostatic Interaction for Epoxy Composites of High Thermal Conductivity and Electrical Resistivity, *Macromol Res* **26** (2018) 521–528.
 15. **M.C. Vu**, Y.H. Bae, M.J. Yu, W. Choi, A. Islam, S. Kim, Thermally conductive adhesives from covalent- bonding of reduced graphene oxide to acrylic copolymer, *J Adhes* **95** (2018) 887–910.
 16. **M.C. Vu**, G.D. Park, Y.H. Bae, M.J. Yu, T.K. An, S.G. Lee, S.R. Kim, Pressure-sensitive adhesive composites with a hydrophobic form of graphene oxide for enhanced thermal conductivity, *Macromol Res* **24** (2016) 1070–1076.
 17. **M.C. Vu**, G. Park, Y. Bae, S. Kim, Enhanced Thermal Conductivity of Pressure Sensitive Adhesives Using Hybrid Fillers of SiC Microparticle and SiC Nanoparticle Grafted Graphene Oxide, *Polymer (Korea)* **40** (2016) 804–812.

Contributing Author Publications

1. Ritesh Kumar, **Minh Canh Vu**, Peiyuan Ma, Chibueze V Amanchukwu, Electrolytomics: A unified big data approach for electrolyte design and discovery, *Chemistry of Materials* (2025), 37, 2720–2734. (Q1, H-index 435, Ranking Top 10% in Chemistry, IF 8.6, 13 citations)
2. P. Mirmira, E. Doyle, P. Ma, A. Garcia, Z. Umlauf, **M.C. Vu**, C.V. Amanchukwu, In situ inorganic and polymer synthesis for conformal hybrid sulfide-type solid state electrolytes, *Chemistry of Materials* (2025), 37, 1069–

1082. (Q1, H-index 435, Ranking Top 10% in Chemistry, IF 8.6, 1 citation)
3. P. Ma, C. Xue, K.H. Wang, P. Mirmira, O. Rivera, **M.C. Vu**, C.V. Amanchukwu, Molecular Structure Optimization of Fluorinated Ether Electrolyte for All Temperature Fast Charging Lithium-Ion Battery, *ACS Energy Letters*, **9** (2025) 6144-6152. (Q1, H-index 223, Ranking 2/124 in Chemistry, IF 20.8, 10 citations)
 4. P. Ma, U. Le, K.H. Wang, **M.C. Vu**, C.V. Amanchukwu, Probing the Influence of Steric Hindrance in Nonfluorinated Ether Electrolytes for Lithium Metal Batteries, *J Electrochem Soc*, **171** (2025) 120536. (Q2, H-index 346, Ranking 29/64 in Chemistry, IF 3.3, 5 citations)
 5. D. Mani, S. Anand, **M.C. Vu**, M.A. Islam, S.R. Kim, Janus polyurethane composite film with highly aligned graphene fluoride and liquid metal: A versatile technique for thermal management and electromagnetic interference shielding, *Compos Part A Appl Sci Manuf* **182** (2024) 108179. (Q1, H-index 227, Ranking 13/130 in Ceramics and Composites, IF 8.9, 15 citations)
 6. K. Pagar, V. Gadore, S.R. Mishra, M. Ahmaruzzaman, P. Basnet, D. Sanap, **M.C. Vu**, K.Y.A. Lin, B. Ravindran, S. Ghotekar, Bio-inspired Sustainable Fabrication of CdO Nanoparticles Using Citrus sinensis Peel Extract for Photocatalytic Degradation of Rhodamine B Dye, *Top Catal* **67** (2024) 1169-1182.
 7. E.S. Doyle, P. Mirmira, P. Ma, **M.C. Vu**, T. Hixson-Wells, R. Kumar, C. V. Amanchukwu, Phase Morphology Dependence of Ionic Conductivity and Oxidative Stability in Fluorinated Ether Solid-State Electrolytes, *Chemistry of Materials* **36** (2024) 5063-5076. (Q1, H-index 435, Ranking Top 10% in Chemistry, IF 8.6, 7 citations)
 8. J.B. Kim, **M.C. Vu**, B.K. Seo, D. Mani, S. Anand, T.H. Jeong, S.R. Kim, Multilayered graphene fluoride and Ti3C2Tx MXene-based aramid nanofiber films with excellent thermal conductivity and electromagnetic interference shielding performance, *Colloids Surf A Physicochem Eng Asp* **685** (2024) 133121. (Q1, IF 5.4, 22 citations)
 9. P. Mirmira, C. Fuschi, Z. Umlauf, P. Ma, E.S. Doyle, **M.C. Vu**, C. V. Amanchukwu, Impact of Processing Methodology on the Performance of Hybrid Sulfide-Polymer Solid State Electrolytes for Lithium Metal Batteries, *J Electrochem Soc* **171** (2024) 030508. (Q2, H-index 346, Ranking 29/64 in Chemistry, IF 3.3, 3 citations)
 10. S. Anand, **M.C. Vu**, D. Mani, J.B. Kim, T.H. Jeong, W.K. Choi, J.C. Won, S.R. Kim, A continuous interfacial bridging approach to fabricate ultrastrong hydroxylated carbon nanotubes intercalated MXene films with superior electromagnetic interference shielding and thermal dissipating properties, *Adv Compos Hybrid Mater* **7** (2024) 33. (Q1, H-index 88, Ranking 2/167 in Polymers and Plastics)
 11. D. Mani, **M.C. Vu**, C.S. Lim, J.B. Kim, T.H. Jeong, H.J. Kim, M.A. Islam, J.H. Lim, K.M. Kim, S.R. Kim, Stretching induced alignment of graphene nanoplatelets in polyurethane films for superior in-plane thermal conductivity and electromagnetic interference shielding, *Carbon* **201** (2023) 568-576. (Q1, H-index 346, Ranking 19/460 in Chemistry, IF 11.2, 76 citations)
 12. S. Anand, **M.C. Vu**, D. Mani, J.B. Kim, T.H. Jeong, M.A. Islam, S.R. Kim, Dual 3D networks of graphene derivatives based polydimethylsiloxane composites for electrical insulating electronic packaging materials with outstanding electromagnetic interference shielding and thermal dissipation performances, *Chemical Engineering Journal* **462** (2023) 142017. (Q1, H-index 337, Ranking 7/274 in Chemical Engineering, IF 13.2, 109 citations)
 13. M. Suganya, S. Anand, D. Mani, **M. Canh Vu**, S. Muniyappan, K.M. Racik, S. Nandhini, J.K. Kumar, Fabrication of novel M-Type cobalt doped barium hexaferrite nanoplatelets /graphitic carbon nitride composite for efficient supercapacitor applications, *Synth Met* **295** (2023) 117341.
 14. C.J. Prabagar, S. Anand, **M.C. Vu**, M.S. Tran, D. Mani, M.M. Martina, S. Pauline, Thermally insulating carbon nanotubes and copper ferrite based porous polydimethylsiloxane foams for absorption-dominant electromagnetic interference shielding performance, *Composites Communications* **42** (2023). (Q1, H-index 73, Ranking Top 10 Polymers and Plastics, 26 citations)
 15. S. Kim, **C.M. Vu**, S. Kim, I. In, J. Lee, Improved Mechanical Strength of Dicatechol Crosslinked MXene Films for Electromagnetic Interference Shielding Performance, *Nanomaterials* **13** (2023) 787.
 16. N.A.T. Thieu, D. Mani, J.C. Won, K.Y. Yon, **M.C. Vu**, S.R. Kim, Vertically interconnected network of graphene fluoride for highly thermoconductive and electrically insulating epoxy composites, *Polym Compos* **43** (2022) 8050-8059. (Q1, H-index 106)
 17. S. Pansambal, A. Roy, H.E.A. Mohamed, R. Oza, **C.M. Vu**, A. Marzban, A. Chauhan, S. Ghotekar, H.C.A. Murthy, Recent Developments on Magnetically Separable Ferrite-Based Nanomaterials for Removal of Environmental Pollutants, *J Nanomater* **2022** (2022) 8560069.
 18. D. Mani, **M.C. Vu**, T.H. Jeong, J.B. Kim, C.S. Lim, J.H. Lim, K.M. Kim, S.R. Kim, 3D structured graphene fluoride-based epoxy composites with high thermal conductivity and electrical insulation, *Compos Part A Appl Sci Manuf*

- 149 (2021) 106585. (Q1, H-index 227, Ranking 13/130 in Ceramics and Composites, IF 8.9, 70 citations,)
19. N.A.T. Thieu, **M.C. Vu**, D.H. Kim, W.K. Choi, S. Kim, Effect of aspect ratio of vertically aligned copper nanowires in the presence of cellulose nanofibers on the thermal conductivity of epoxy composites, *Polym Adv Technol* **31** (2020) 2351.
 18. V.C. Doan, **M.C. Vu**, N.A.T. Thieu, M.A. Islam, P.J. Park, S.R. Kim, Copper flake-coated cellulose scaffold to construct segregated network for enhancing thermal conductivity of epoxy composites, *Compos B Eng* **165** (2019) 772–778. (39 citations, Q1, H-index 227, Ranking 11/720 in Mechanical Engineering, IF 14.2, 39 citations)
 20. C. Vu, M.J. Yu, **M.C. Vu**, S. Han, J.H. Park, S. Kim, Fabrication and characterization of the nano- and micro-particles applied dry adhesives, *Journal of Adhesion and Interface* **20** (2019) 23–28.
 21. V.C. Doan, **M.C. Vu**, M.A. Islam, S.R. Kim, Poly(methyl methacrylate)-functionalized reduced graphene oxide-based core-shell structured beads for thermally conductive epoxy composites, *J Appl Polym Sci* **136** (2019) 47377. (Q2, H-index 200)
 22. N.A.T. Thieu, **M.C. Vu**, E.S. Lee, V.C. Doan, S.R. Kim, Enhancement of Thermal Conductivity of Poly(methylmethacrylate) Composites at Low Loading of Copper Nanowires, *Macromol Res* **27** (2019) 1117–1123.
 23. Y.H. Bae, G.D. Park, H.O. Jung, **M.C. Vu**, S.R. Kim, Thermal conductivity improvement by Cu surface treatments and incorporation of pmma beads on Cu/epoxy composites, *Polymer (Korea)* **40** (2016) 148–153.
 24. Y.H. Bae, M.J. Yu, **M.C. Vu**, W.K. Choi, S.R. Kim, Synergistic effects of segregated network by polymethylmethacrylate beads and sintering of copper nanoparticles on thermal and electrical properties of epoxy composites, *Compos Sci Technol* **155** (2018) 144–150.
 25. M.J. Yu, T.S. Kwon, Y.H. Bae, **M.C. Vu**, B.C. Lee, S.R. Kim, Effects of carbon-based nanofillers on the structure and property of phenolic foam, *Polymer (Korea)* **42** (2018) 133–139.

Book Chapter

1. **M.C. Vu**, S. Ghotekar, T.-D. Pham, N.M. Viet, R. Oza, M. Bilal, A. Roy, Plant-Mediated Eco-Friendly Synthesis of Platinum Nanoparticles and Their Applications, *Plant and Nanoparticles* **6** (2022) 155–170.
2. **M.C. Vu**, S. Ghotekar, N.M. Viet, H. Dabhane, R. Oza, A. Roy, Green Synthesis of Plant-Assisted Manganese-Based Nanoparticles and Their Various Applications, *Plant and Nanoparticles* (2022) 339–354.
3. Y. Abhale, T. Pagar, R. Oza, K.-Y.A. Lin, A.P. Larios, **C.M. Vu**, S. Ghotekar, Green and Cost-Effective Nanomaterials Synthesis from Aquatic Plants and Their Applications, *Secondary Metabolites Based Green Synthesis of Nanomaterials and Their Applications*, (2023) 309–325.

Scholarships and Fellowships

University of Chicago Postdoctoral Fellowship	2021
Korea National University Postdoctoral Fellowship	2021
Korea National University Postgraduate student scholarship	2017
Korean Government Scholarship for outstanding postgraduate students	2015

Awards and Honors

5 th International Conference on Emerging Advanced Nanomaterials, Travel Grant	2024
Best Poster Award at The 11 th International Workshop on Advanced Materials Science and Nanotech	2024
Best Oral Presentation at Autuum Materials Research Society (MRS) conference	2023
Pritzker School of Molecular Engineering of Excellence in Mentoring Award	2023
Vietnamese Golden Young Scientist Award, Finalist	2021
Korea National University of Transportation Outstanding Thesis Award	2020
Best Oral Presentation at Spring Polymer Society Korea conference	2019
Best Poster Presentation at European Materials Research Society (EMRS)	2018
Best Poster Presentation at Autumn Polymer Society Korea conference	2017
Best Poster Presentation at Autumn Adhesive Society Korea conference	2016

Teaching Experience

Industrial University of Ho Chi Minh City – Visiting Professor	Ho Chi Minh City, Vietnam
Materials Science and Engineering: An Introduction	Aug. 2024 – Dec. 2024
University of Chicago – Lecturer	Chicago, USA
MENG 24100, Molecular Engineering Thermodynamics of Phase Equilibria	Aug. 2022 – Dec. 2023
MENG 25310, Energy Storage and Conversion Devices	Aug. 2012 – Dec. 2023
University of Da Nang – Visiting Lecturer,	Da Nang, Viet Nam

Polymer Chemistry
Electrochemistry
Korea National University– Teaching Assistant,
Introduction to Polymeric Materials
New Trends in Polymers
Polymer Composites
Display Materials
Structural and Properties of Polymer

Mar. 2021 – Dec. 2024
Mar. 2021 – Mar. 2024
Chungju, South of Korea
Mar. 2017 – May 2020
Mar. 2017 – May 2020
Mar. 2017 – May 2020
Mar. 2017 – May 2020
Mar. 2017 – May 2020

Conferences

Invited Speaker

- ♦ **2025:** 4th International Conference on Polymer Science and Engineering, Houston, USA.
- ♦ **2024:** 5th International Conference on Emerging Advanced Nanomaterials ICEAN 2024, Newcastle, Australia
- ♦ **2024:** 7th International Conference on Composite Material, Polymer Science, and Engineering (CMPSE2024), Bangkok, Thailand
- ♦ **2024:** International Conference on Polymers Science and Engineering, Kuala Lumpur, Malaysia
- ♦ **2023:** 2nd International Conference on Graphene & 2D Materials, London, UK
- ♦ **2023:** 12th International Conference on Advanced Materials and Engineering Materials (ICAMEM2023), Bangkok, Thailand
- ♦ **2023:** 5th International Conference on Metal Material Processes and Manufacturing (ICMMPM2023), Jeju, South Korea

Oral Presentation

- ♦ **2024:** 1st International Conference on Advances in Carbon Science and Innovation (ICACSI), Sydney, Australia.
- ♦ **2023:** Fall meeting, **Materials Research Society** (MRS), Boston, USA.
- ♦ **2022:** Fall meeting, **American Chemical Society** (ACS), San Francisco, USA.
- ♦ **2020:** Spring The **Polymer Society Korea**, Deajeon, South Korea.
- ♦ **2018:** Fall meeting, **European-MRS**, Warsaw, Poland.

Poster Presentation

- ♦ **2020:** Fall The Polymer Society Korea, Jeju, South Korea.
- ♦ **2019:** Fall The Polymer Society Korea, Jeju, South Korea.
- ♦ **2019:** Spring The Polymer Society Korea, Deajeon, South Korea.
- ♦ **2018:** Fall The Polymer Society Korea, Jeju, South Korea.
- ♦ **2018:** Spring The Polymer Society Korea, Deajeon, South Korea.
- ♦ **2017:** Fall The Polymer Society Korea, Jeju, South Korea.
- ♦ **2017:** Spring The Polymer Society Korea, Deajeon, South Korea.
- ♦ **2016:** Fall The Polymer Society Korea, Jeju, South Korea.

Contributed Reviews

- ♦ **2025:** Materials today communications (10), Ceramics International (1), Composites Science and Technology (1), Composite Part B (3), Computational materials science (2), Construction & building materials (1), International journal of biological macromolecules (1), Journal of alloys and compounds (1), Surfaces and interfaces (3), Polymer testing (1), Nature communications (2), Nano energy (1), Materials today chemistry (1).
- ♦ **2024:** Materials Research Bulletin, Journal of Alloys and Compounds (5), Journal of Colloids and Surfaces (2), Ceramics International (2), Advanced Engineering Materials (2).
- ♦ **2023:** ACS Nano, Chemical Engineering Journal, Journal of Materials Chemistry A.
- ♦ **2022:** Journal of Applied Polymer Science (7), ACS Applied Polymer Materials (2), ACS Applied Electronic Materials.
- ♦ **2021:** Journal of Applied Polymer Science (6), ACS Applied Polymer Materials, ACS Applied Electronic Materials.
- ♦ **2020:** ACS Applied Materials and Interfaces
- ♦ **2019:** ACS Applied Materials and Interfaces